

```
</PackageSelection>
<LanguageSelection default_language="en_US">
</Language>en_US</Language>
</LanguageSelection>
</PardusmanProject>
```

At the end of the wizard, the config file is generated. The project_file is designed as a tarball (.tar.gz) with the following structure:

```
Project_file.tar.gz
---+ user_contents.tar.gz ( user home contents )
---+ user_presentation.odp
---+ RELEASE.txt
---+ project.xml
---+ user_wallpapers (wallpaper)
---+ wallpaper1.png
```

On generating the project file, it is sent to the Buildfarm Process Queue. Buildfarm is a component of Pardusman that performs the distro builds from the given project configuration file. Buildfarm is run as a daemon that is run all the time. Process Queue maintains the list of distro build requests from users along with the link of the project file generated. There is another queue called the On_Progress queue, which maintains the list of distro projects for which the builds are in progress.

Distro building is a CPU-intensive process and it requires a heavy hardware configuration to build multiple distros at a time. On my laptop, performing more than two distro builds raises temperatures to very high levels and the laptop gets powered off. So simultaneous multiple distro builds are to be restricted according to the server capacity. Buildfarm is provided with a configuration file through which lots of parameters can be specified. It includes BUILD_LIMIT = No of builds to be performed at a time.

The Buildfarm_queue is capable of holding any number of user build requests. The On_Progress queue is restricted according to BUILD_LIMIT. Once the build of one project is completed, a project is fetched from the Buildfarm_queue to the On_Progress queue and the build will be started.

Users can keep track of the progress of their project build requests through the Userlog page. In the Userlog page, once the user requests for a build and the project is sent to the Buildfarm_Queue, it will be shown as 'scheduled' against the line that corresponds to a project build. Once the project is in the build state and is inside the On_Progress queue, it will be shown as 'in progress'. If the project build is successful, it will be shown as 'Completed' and if it fails, will show up as 'Failed'. Once the build is completed, a download link to the image will be provided. It also will list out a Logfile link, which can be used for debugging if the project build fails and can send a bug report or *dugg* with the reasons for it.

There are lots of complex things underlying the Buildfarm component. Explaining those implementation details is not within the scope of this article. Still, I would like to share some of the best features that come with the *pisi* package management tool, which helped me automate the build of the filesystem base easily.

For example, if you want to build a filesystem base in the current directory 'pardus-root', perform the following:

```
# pisi --yes-all -D' /pardus-root' ar pardus-install http://paketiier.pardus.org.tr/pardus-2009/pisi-index.xml.bz2
```

It initiates the directory 'pardus-root' as the root filesystem attached with a repository.

Now we can install packages to this filesystem as:

```
# pisi --yes-all --ignore-comar -D' /pardus-root' if -c system_base
# pisi --yes-all --ignore-comar -D' /pardus-root' if <package_name>
```

Did you find any package management tool that is capable of this? Apt, Yum, Emerge...? There are more. Suppose I have a bunch of *pisi* package files in a directory, I can make that directory a *pisi* repository. What do you think?

Change the current directory to the above mentioned directory and execute the following:

```
# pisi lx
```

It will create the index file pisi-index.xml and pisi-index.xml.bz2

Now you can add the directory as the repository, using:

```
#pisi ar 'repo-name' ar <directory_path/pisi-index.xml.bz2>
```

That is all about the essentials of Pardusman. We are setting up Pardusman on the official Pardus server for all of you to master your own custom Pardus builds. Keep yourself updated with <http://pardusman.pardus.org.tr>.

I have already set up the Pardusman wizard right there. The Buildfarm component is not yet initiated as the daemon. The Pardus team will run the Buildfarm soon. Hence you can have your own distro versions of a custom-built Pardus, with your own wallpaper, packages, home folder contents and a lot more.

Grab the code at http://google-summer-of-code-2009-pardus.googlecode.com/files/Sarath_Lakshman.tar.gz

Enjoy Pardusman. Get, set and burn!

Happy hacking! 

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